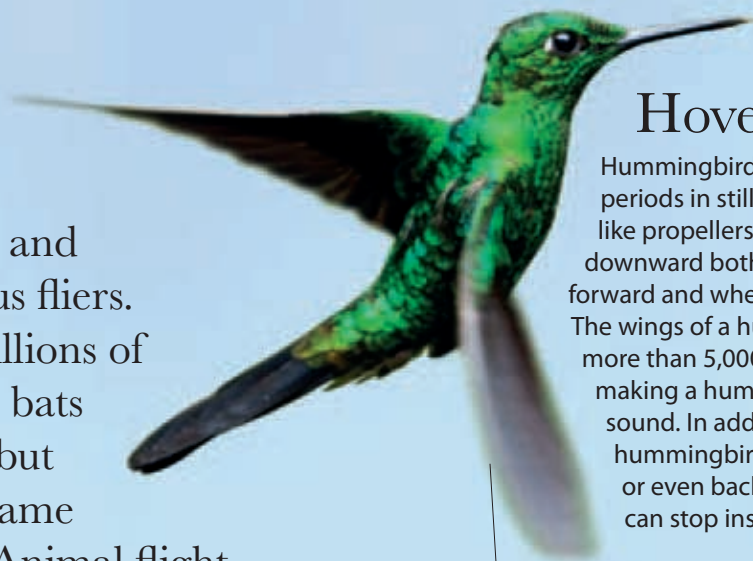


Flying

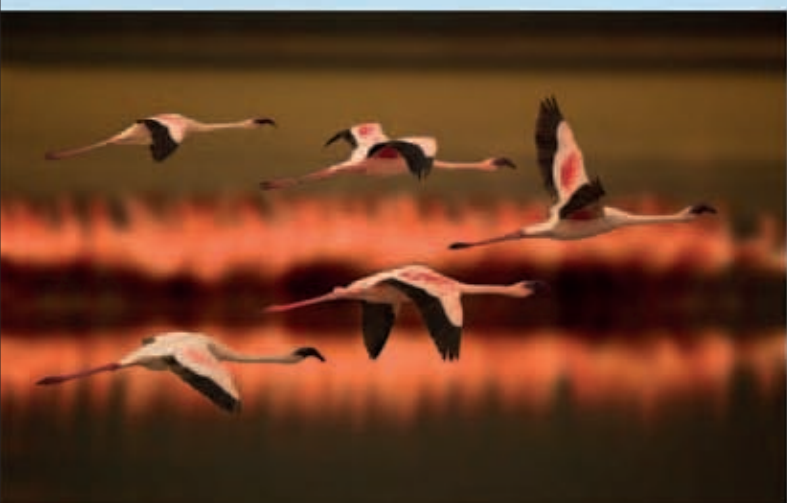
Many animals can glide, but only birds, bats, and insects can flap their wings to stay in the air. Birds fly farthest and fastest, but insects are the most numerous fliers. Migrating swarms sometimes contain billions of insects, taking days to pass by. Birds and bats have wings made from feathers or skin, but insects' wings are made of chitin—the same substance that forms their exoskeleton. Animal flight times vary enormously. Ladybugs touch down after a few minutes, but some sea birds can stay airborne for several years before they land to breed.



Hovering

Hummingbirds can hover for long periods in still air. Their wings work like propellers, pushing the air downward both when they sweep forward and when they sweep back. The wings of a hummingbird can beat more than 5,000 times a minute, making a humming or buzzing sound. In addition to hovering, hummingbirds can fly sideways or even backward, and they can stop instantly in midair.

Wing sweeping back, pushing air downward



Feathered fliers

With their outstretched legs and necks, greater flamingos look supremely elegant as they cruise through the air. Like most birds, they have lots of special features that enable them to fly. Their skeletons are extra-light, with bones that are hollow or paper-thin, but their wing muscles are huge, making up more than one-third of their body weight. Their lungs are highly efficient, making sure that their muscles get the maximum amount of oxygen. Finally, some feathers keep them warm and streamlined, and others provide the upward lift that keeps them in the air.



Elytrum fixed in place as soon as ladybug is airborne

Inside a wing

A bird's wing bends at three joints, equivalent to the human shoulder, elbow, and wrist. The power behind a wing beat comes mainly from the pectoral, or breast, muscles. Other muscles adjust the wing's shape in flight, or fold it up.

